

Amazon Interview Experience

About the company:

Amazon is a leading American multinational technology company with a primary focus on e-commerce, cloud computing, online advertising, digital streaming, and artificial intelligence. It is recognized as one of the "Big Five" tech giants in the U.S. As a product-driven company, Amazon is also part of the well-known MAANG group, representing the most influential companies in the tech industry.

- **Role Offered:** SDE-2m Intern.
- **Stipend:** 1,10,000 INR + 400 USD per month.
- **Selection Process:** PPT, Online Assessment, Technical Interview.

Online Test:

The assessment included two coding questions and a work personality test. The online Assessment was conducted in HackerRank platform.

- **Coding Questions:** 2 (Total time: 70 minutes)
- **Personality Test:** No time limit (Mandatory to complete)

Coding Question 1:

Given an integer x and a vector `cost` of integers. Your task is to count the number of pairs (i, j) where $i < j$ and the sum of `cost[i] + cost[j]` is divisible by x .

Approach:

I used modulo arithmetic to find pairs whose sum is divisible by a given integer x . By calculating the remainder of each element in the array and using a hash map to track the frequency of these remainders, we can efficiently find complement remainders that, when added to the current element's remainder, result in a multiple of x . Then I counted the valid pairs by incrementing based on the frequency of the complements.

Coding question 2:

Given a list of strings patterns, each consisting of characters and wildcard symbols (?), determine the minimum number of columns where a character conflict occurs. A character conflict in a column happens when different non-wildcard characters are found in that column. Wildcard symbols are ignored in conflict checks. Return the count of such conflicting columns.

Approach:

For each column, we shall track characters seen so far, ignoring wildcard characters (?). If two non-wildcard characters differ within the same column, it indicates a conflict, and we increment the count of such conflicts.

Based on this Online Assessment, 52 people were shortlisted for the next round.

Technical Interview:

The interview took place on Amazon's Chime platform, with a link provided to join the meeting. Two recruiters participated: one as the primary interviewer and the other as an observer. Each of us introduced ourselves, and it was noted that there would be no separate HR round. So make sure that you showcase your technical strengths and relevant projects during your introduction. Following this, you'll be asked to share your screen and use LiveCode to tackle the coding questions.

The interviewer will post the coding question on the platform. Be sure to read and understand the question thoroughly. It's a good idea to take a few minutes to consider your approach before starting. If you have any questions about the problem, don't hesitate to ask the interviewer for clarification.

Begin by describing the brute force approach before moving on to the optimized solution. Ensure you engage with the interviewer during the coding process and avoid being silent. When discussing time and space complexities, provide a thorough explanation.

Coding Question 1:

Given a 2D array where each index represents a bomb with a specific detonation radius. When a bomb detonates, it can set off other bombs within its radius. The task is to find the maximum number of bombs that can be detonated starting from any single bomb, considering the cascading effect of detonation.

Problem Link: <https://leetcode.com/problems/detonate-the-maximum-bombs/description/>

First, the interviewer asked me to explain the brute force approach, which I did using DFS. Then, she asked about the time and space complexities. After 5 to 10 minutes, she asked me to code the function. She also wanted to know if there were any possible optimizations. I thought about it for a few minutes, and then she moved on to the next question.

Coding Question 2:

Given an array of integers where each integer represents a ball of a specific color: 0 for white, 1 for red, and 2 for blue. The task is to sort this array in-place such that all balls of the same color are grouped together. This problem is often referred to as the "Dutch National Flag Problem."

Problem Link: <https://www.geeksforgeeks.org/sort-an-array-of-0s-1s-and-2s/>

She directly asked for the optimized approach. Since I was familiar with the three-pointer algorithm to solve the problem, I explained that method to her. I was then asked to code the solution and perform a dry run with the provided sample input. Finally, she inquired about the time and space complexities of the code.

The interview was scheduled to last exactly one hour. You are likely to be selected if you solve both questions within the time limit. At the end, the interviewer asked if I had any questions for her.

Be confident and explain your approach clearly. The interviewers are mostly interested in your thought process rather than your code.

All the very best!

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